

The Memory Runtime Hypothesis

A Repo-Native Theory of Persistent Cognitive Infrastructure for NeuroAI

Series: *Repo-Native NeuroAI Infrastructure*

Paper: I of IV

Packet ID: NEUROAI-REPO-CERN-EXPANDED-20260601

Authority state: CLAIM_HARDENED / PUBLIC_SAFE_DRAFT

Primary substrate: R1, R11, R12, R13 with NeuroAI context from E1, E2, E5

Claim posture: supported architecture plus C9 theorem candidate

Boundary: no clinical validation, biological equivalence, production deployment, external notarization, legal filing, or CERN affiliation is claimed.

Next proof move: run longitudinal memory, authority, denial-path, replay, poisoning, and rollback tests against less separated agent architectures.

Abstract

A system that cannot preserve the authority of its own memory cannot yet be trusted as a research agent. It may answer fluently, retrieve fragments, write reports, and even appear coherent inside a single session, but cognition that cannot distinguish draft from canon, transient synthesis from durable memory, and heavy computation from lawful update remains structurally fragile. This paper introduces the **Memory Runtime Hypothesis**: long-horizon NeuroAI agents require a protected memory-governance substrate in which active cognition, canonical memory, heavy synthesis, and runtime authority are separated into explicit layers. Medina's repo intelligence supplies the architectural seed for this hypothesis through NOVA_ROOT, MEMORY_RUNTIME, ACTIVE_STATE_RUNTIME, and FOUNDATION_FLOOR. The claim is not that Medina has built a brain-equivalent system or proven a theorem of cognition. The claim is sharper and safer: the repo substrate defines a software-level cognitive infrastructure hypothesis that can be tested against agentic failure modes including memory corruption, claim drift, authority confusion, and false promotion of hypotheses into facts.

1. The Forgetting Machine

Most contemporary AI systems remain brilliant forgetting machines. They can solve a problem, narrate a plan, or synthesize a document, but their continuity is often carried by prompt context, retrieval glue, or external orchestration rather than by an explicit cognitive law of memory. The result is a strange form of intelligence: capable in the moment, uncertain across time. It can generate an answer, but it may not know whether the answer is a proposal, a verified result, a private note, a public claim, or a canonical memory update.

For ordinary chat, that may be tolerable. For research agents, it is not. A research agent must remember not only information but the authority of information. It must know which statements are hypotheses, which are supported by evidence, which are internal, which are public-safe, which are blocked, and which have been promoted through review. If that distinction collapses, the agent does not merely make a factual mistake. It corrupts the epistemic chain by which future work decides what may be built, published, protected, deployed, or notarized.

This is where Medina's repo intelligence becomes NeuroAI-relevant. The repo does not present a biological neural model. It presents a memory-governed architecture for synthetic cognition. In the Alpha-family runtime contracts, NOVA_ROOT governs lawful authority, MEMORY_RUNTIME owns canonical append-only memory, ACTIVE_STATE_RUNTIME performs live work and synthesis, and FOUNDATION_FLOOR handles heavy compute while explicitly lacking direct canonical write authority. That separation is not decorative. It is the beginning of a research hypothesis: durable intelligence requires runtime memory law.

Claim Boundary Box 1

Supported by repo substrate: the architecture and its boundary language exist in the supplied runtime contracts and THESIS ledgers.

Hypothesis: separated memory authority improves long-horizon agent reliability.

Missing proof: denial-path tests, replay tests, poisoning tests, longitudinal benchmarks, and comparative architectures.

Blocked claim: Medina has proven a brain-equivalent runtime or deployed a fully enforced NeuroAI memory system.

2. NeuroAI Beyond Neural Mimicry

NeuroAI is often introduced through the question of how neuroscience can inform artificial intelligence. That question is real, but too often it narrows into a search for visible resemblance: spiking units, recurrent loops, embodied agents, cortical analogies, or neural predictivity benchmarks. Those are valuable directions. The 2023 NeuroAI position paper argues that next-generation AI may need deeper engagement with biological intelligence, including embodied evaluation and an understanding of which brain details matter for intelligence. Brain-Score and NeuralBench show the field’s movement toward benchmarkable comparison between artificial models and neural or behavioral data.

Yet there is another layer of NeuroAI that is less glamorous and more fundamental: cognitive infrastructure. Brains are not only prediction machines. They are regulated memory systems. They preserve, consolidate, inhibit, retrieve, update, and contextualize. They maintain distinctions between present action and long-term identity. They operate under metabolic and developmental constraints. They are not reducible to a single feedforward response.

Medina’s runtime architecture enters here as a software-level hypothesis, not a biological claim. The analogy is functional and bounded: an artificial cognitive system may need protected memory, active working state, authority gates, and slower deliberative synthesis lanes. This is not the hippocampus, cortex, basal ganglia, or thalamus in disguise. It is not a neuroscientific mapping. It is a claim about artificial cognition under conditions of time, revision, proof, and publication.

The phrase “brain-like” should therefore be avoided unless tied to a specific benchmark. Medina’s safer phrase is **memory-preserving artificial cognition**. That term does not pretend to solve neuroscience. It states the engineering and research problem: can an agent preserve the authority, lineage, and boundary state of its own knowledge while acting across many tasks and revisions?

3. Runtime Before Reasoning

The Memory Runtime Hypothesis begins before any particular reasoning algorithm. It says the substrate matters. If a system lacks protected memory law, then reasoning can become a transient performance rather than a cumulative cognitive act.

The repo-defined stack can be read as four interacting layers.

Layer	Repo Role	Cognitive Interpretation	Claim Boundary
NOVA_ROOT	lawful runtime authority	global constraint on what may be promoted	architecture exists as doctrine; enforcement proof not supplied
MEMORY_RUNTIME	canonical append-only memory	durable lineage and identity substrate	memory law is specified; implementation proof remains bounded
ACTIVE_STATE_RUNTIME	live synthesis and decisions	working cognition and action surface	live work exists as process; not proof of safe autonomy
FOUNDATION_FLOOR	heavy compute lane	slow deliberation, batch synthesis, prior-art search, stress testing	may not write canonical memory directly; denial-path enforcement is unproven

This table is the technical object of Paper I. Its importance is not that it imitates a neural system. Its importance is that it refuses the most dangerous simplification in agent design: allowing the same process that imagines to also canonize. A system that can generate hypotheses should not automatically make them memory. A system that can perform heavy synthesis should not automatically rewrite lineage. A system that can produce public text should not automatically define public truth.

The THESIS proof ledger makes the limit clear. Packet tooling, hash builders, render helpers, and local hash manifests have proof support in prior packet work. But denial-path enforcement that Foundation Floor cannot

write canonical memory directly remains not proof-backed. Comparative auditability gains from compute/memory separation also remain unproven. That is precisely the right posture for a serious paper: the architecture is strong enough to become a research program, not strong enough to become a victory claim.

4. Memory Is Not a Log

A common mistake in agent architecture is to treat memory as storage. A vector database, transcript archive, or append-only log may preserve data, but it does not automatically preserve authority. Memory for a research agent must carry at least five states: source identity, claim class, evidence posture, release boundary, and promotion history.

This is where Medina’s THESIS machinery enriches the runtime hypothesis. THESIS packets require source artifact identity, authority state, evidence state, claim-class summary, provenance references, policy boundary, derivative policy, review state, promotion state, and destination surface. A memory entry without those fields can become a dangerous fossil. It may be retrieved later without knowing whether it was private, disproven, hypothetical, or superseded.

In biological language, one might be tempted to compare this to consolidation, executive control, or metacognition. Those comparisons can help generate experiments, but they should not be used as proof. The stronger framing is infrastructural: memory is not a log; memory is governed continuity.

A governed memory substrate should answer:

1. Who or what produced this claim?
2. What evidence supports it?
3. What class of claim is it?
4. What release surface may contain it?
5. What revision supersedes it?
6. What proof gap prevents promotion?
7. What downstream packet may inherit it?

If a system cannot answer those questions, its memory is merely accumulation. Accumulation is not cognition.

5. Authority as a Cognitive Constraint

The term “authority” can sound administrative, but in this architecture authority is cognitive. It determines what kind of operation a process is allowed to perform on knowledge. A hallucination problem is not only a generation problem. It is also an authority problem: the system speaks as if a sentence has more status than it deserves.

The bounded authority membrane theorem candidate can be stated as follows:

Theorem Candidate: Bounded Authority Membrane

In long-horizon artificial research agents, auditability and epistemic stability improve when high-capacity generative or synthesis processes cannot directly mutate canonical memory, public claims, or deployment state without mediated review.

This is a theorem candidate, not a proven theorem. It predicts that separated architectures should outperform fused architectures on longitudinal coherence, contradiction control, rollback recovery, and release-boundary preservation. It also predicts costs: slower promotion, more review overhead, and possible friction for creative exploration. Those costs must be measured.

The key NeuroAI insight is that cognition is not merely capability; it is capability under constraint. A system’s intelligence is shaped by what it is prevented from doing as much as by what it can generate.

6. The Brain Analogy Stops Here

This paper deliberately refuses a strong biological analogy. Medina’s runtime stack is not a validated model of hippocampus, cortex, basal ganglia, or prefrontal control. It is not evidence of consciousness, sentience, clinical cognition, or human-like memory. The architecture is a software research object.

Still, NeuroAI does not require every contribution to be a direct biological mapping. The field also needs artificial systems that make cognitive questions testable: how memory is preserved, how action is authorized, how uncertainty is bounded, how claims become durable, and how agents recover from drift. Medina’s runtime stack belongs in that class. It is a proposed instrument for studying memory-preserving artificial cognition.

External literature supports only the seriousness of the research direction. NeuroAI calls for brain-informed, embodied, robust, and benchmarkable AI. Predictive coding and active inference provide one vocabulary for action and perception, but reviews caution that their empirical status remains uneven and that model comparison is essential. Medina should inherit that caution. It should not use NeuroAI language to promote its architecture beyond evidence.

7. Predictions and Experiments

The Memory Runtime Hypothesis becomes scientific only when it generates tests. The following experiments would begin to promote it from architecture thesis to supported result.

Experiment	Setup	Expected Advantage If Hypothesis Holds	Failure Signal
Longitudinal claim recall	compare separated vs fused agents over multi-week research tasks	separated agents preserve claim class and revision state better	no difference in claim drift
Authority-boundary recognition	inject private, public, and hypothetical material	separated agents block unsafe promotion more reliably	private material leaks or hypotheses become facts
Denial-path enforcement	attempt direct heavy-compute memory writes	Foundation Floor cannot canonize without mediation	direct canonical mutation succeeds
Replay and poisoning	introduce corrupted historical context	governed memory resists contamination or flags conflict	poisoned claim becomes canonical
Rollback recovery	force incorrect promotion, then provide correction	lineage allows clean rollback and supersession	memory remains contradictory

These experiments should not be hidden in an appendix. They are the path by which the paper earns its seriousness.

8. Objections

Objection 1: This is governance architecture, not NeuroAI.

Response: if NeuroAI is only neural resemblance, the objection holds. But if NeuroAI includes the study of architectures that support memory, action, adaptation, and evaluation, then memory governance is a legitimate research layer. The paper does not claim biological equivalence; it claims cognitive infrastructure relevance.

Objection 2: Separation may reduce creativity.

Response: yes. The architecture may create friction. The correct question is not whether friction exists, but whether it reduces catastrophic memory corruption and false promotion enough to justify its cost.

Objection 3: Authority language may smuggle in institutional values.

Response: all memory systems smuggle in values through what they preserve, forget, and promote. Medina’s advantage is that the value boundary is explicit and testable.

9. Conclusion: From Prompted Intelligence to Remembering Systems

The next step in NeuroAI is not only bigger models, better embeddings, or closer neural benchmarks. It is remembering systems: agents that know what their own outputs are allowed to become. Medina’s repo intelligence offers a concrete hypothesis for such systems through runtime authority, canonical memory, active state, and heavy compute separation. The architecture is not proven. That is why it is a paper. Its value lies in turning a private design instinct into a public, falsifiable research program.

10. A Minimal Formal Model

The Memory Runtime Hypothesis can be expressed without pretending to be a full mathematical theory. Let an agent state be divided into four components: an active working state A , a canonical memory state M , a heavy synthesis process F , and an authority policy P . In a fused architecture, generation may directly update memory: $A \rightarrow M$. In the Medina-style separated architecture, heavy or active synthesis must pass through policy mediation before canonical update: $F \text{ or } A \rightarrow P \rightarrow M$.

The research question is whether the mediated transition improves epistemic stability over long horizons. Stability can be defined as lower contradiction rate, lower unsupported promotion rate, higher revision traceability, higher rollback success, and better preservation of release boundaries. This model is intentionally minimal. It does not assume that P is perfect, that M is immune to corruption, or that separation automatically improves all tasks. It only makes the comparison measurable.

A first experiment would instantiate two agents with identical language and retrieval capabilities but different memory write permissions. Both agents would work through the same evolving research corpus. The fused agent may directly update canonical memory after generating conclusions. The mediated agent must classify claim state, attach evidence, and pass promotion checks before canonical write. The comparison should measure not only final answer quality but memory health after many revisions.

This is the point at which the architecture becomes NeuroAI research rather than doctrine. It proposes a cognitive substrate, defines alternative systems, and names observable outcomes.

References

Repo Intelligence Substrate

- R1. `/workspace/agent_files/alpha-family/runtime-contracts.md` — runtime authority laws for NOVA_ROOT, MEMORY_RUNTIME, ACTIVE_STATE_RUNTIME, and FOUNDATION_FLOOR.
- R2. `/workspace/agent_files/internal-repo/mini-brain.md` — THESIS routing scaffold for multi-surface work.
- R3. `/workspace/agent_files/internal-repo/heavy-and-weights-heart.md` — coherence, continuity, and anti-fragmentation modulation reference.
- R4. `/workspace/agent_files/internal-repo/source-to-forge.md` — doctrine-to-artifact transformation rail.
- R5. `/workspace/agent_files/internal-repo/forge-to-deploy.md` — generated artifact promotion and release-readiness rail.
- R6. `/workspace/agent_files/internal-repo/nexus-registration-flow.md` — shared manifest, contract, and lineage registration rail.
- R7. `/workspace/agent_files/internal-repo/agent-engine-manifest.md` — reusable engine contracts and cross-agent registry objects.
- R8. `/workspace/agent_files/alpha-family/family-manifest.md` — Medina Alpha-family roles, including AURO and THESIS Alpha.
- R9. `/workspace/agent_files/docs/thesis-packet-pipeline-specification.md` — canonical packet pipeline and claim-boundary stages.
- R10. `/workspace/agent_files/docs/packet-stack-taxonomy-research-archive-proof-deployment.md` — packet-stack flows across proof, publication, archive, deployment, and notary prep.
- R11. `/workspace/memory/thesis/claim-ledger.md` — current claim classes and blocked promotion states.
- R12. `/workspace/memory/thesis/proof-ledger.md` — proof posture, current evidence ceiling, and missing verification.
- R13. `/workspace/memory/thesis/release-boundaries.md` — public-safe, internal-only, and blocked release language.

External NeuroAI and Benchmark Literature

- E1. Zador et al., “Catalyzing next-generation Artificial Intelligence through NeuroAI,” *Nature Communications* (2023). <https://www.nature.com/articles/s41467-023-37180-x>
- E2. Brain-Score platform and benchmark program. <https://www.brain-score.org/>
- E3. Banville et al., “NeuralBench: A Unifying Framework to Benchmark NeuroAI Models,” arXiv:2605.08495 (2026). <https://arxiv.org/abs/2605.08495>

- E4. NeuroBench benchmark framework for neuromorphic algorithms and systems, arXiv:2304.04640 (2023). <https://arxiv.org/abs/2304.04640>
- E5. Walsh et al., “The empirical status of predictive coding and active inference,” *Neuroscience & Biobehavioral Reviews* (2024). <https://pubmed.ncbi.nlm.nih.gov/38030100/>
- E6. Hasson, Nastase, and Goldstein, “Direct Fit to Nature,” *Neuron* (2020). <https://pmc.ncbi.nlm.nih.gov/articles/PMC7096172>
- E7. Schuman et al., “Embodied neuromorphic intelligence,” *Nature Communications* (2022). <https://www.nature.com/articles/s41467-022-28487-2>
- E8. “The brain’s unique take on algorithms,” *Nature Communications* (2023). <https://www.nature.com/articles/s41467-023-40535-z>
- E9. Santoro et al., “Cortical encoding of speech enhances task-relevant acoustic information,” *Nature Human Behaviour* (2019). <https://www.nature.com/articles/s41562-019-0648-9>
- E10. “Improving semantic understanding in speech language models via brain-tuning,” arXiv:2410.09230 (2024). <https://arxiv.org/abs/2410.09230>